深圳市金逸晨电子有限公司

ShenZhen GoldenMorning Electronic Co.,Ltd

Model NO 型号	GMT096-04-WG
Product Name 产品名称	0.96 TFT LCD Module
Version 版本号	V1
Date 日期	2022-05-17

☐**Preliminary Specification**（规范草案）

☑**Final Specification**（最终规格）

PREPARED 编辑	**CHECKED** 检查核对	**APPROVED** 审批人

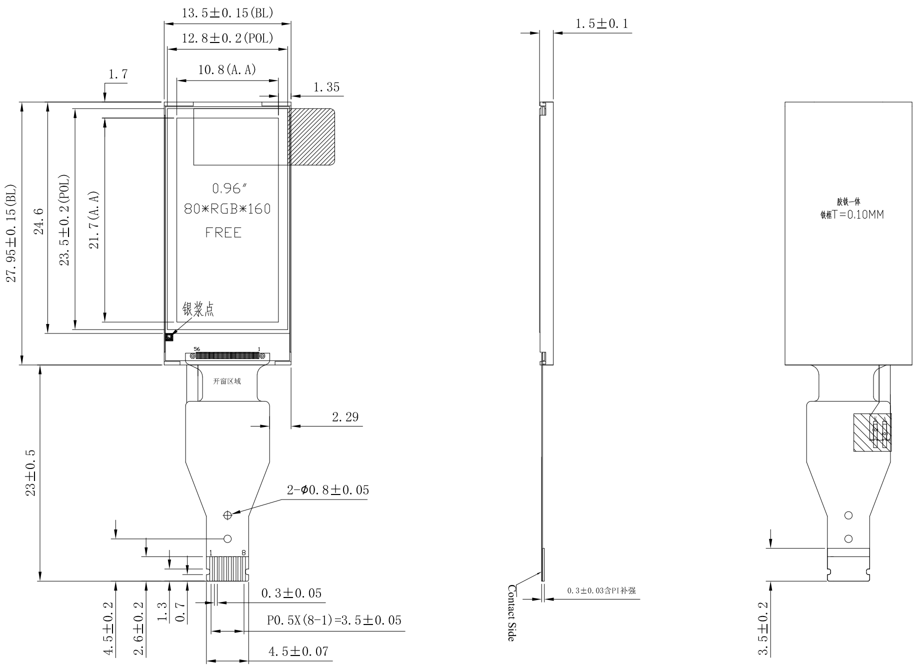
TABLE OF CONTENT

1. General Specifications（概况）

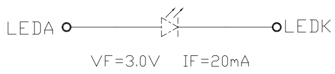
GMT096-04-WG is a 80RGBX160 dot-matrix TFT LCD module. This module is composed of a TFT LCD Panel, driver ICs, FPC and a Backlight unit.

NO.	Item	Contents	Unit
1	LCD Size	0.96inch(Diagonal)	-
2	Display Mode	Normally black	-
3	Resolution	80(H)RGB x 160(V)	-
4	Dot pitch	0.135 (H) x 0.1356(V) mm	-
5	Active area	10.8(H) x 21.7 (V) mm	-
6	Module size	13.5(H) x 27.9(V) x1.5(D) mm	-
7	Color arrangement	RGB Vtertical stripe	-
8	Interface	SPI4 LINE	-
9	Drive IC	ST7735S	-
10	Luminance(cd/m2)	260 (TYP)	
11	Viewing Direction	All View	
12	Backlight	1 White LED	
13	Operating Temp.	-20℃~ + 70℃	℃
14	Storage Temp.	-30℃~+ 80℃	℃
15	Weight	TBD	g

Item	Date	Remark
A	20191114	Original Drawing



NO	SYMBOL
1	LEDA
2	GND
3	RESET
4	RS
5	SDA
6	SCL
7	VDD
8	CS



LED CIRCUIT DIAGRAM:

NOTES:

1. DISPLAY TYPE: 0.96" TFT
2. VIEWING DIRECTION: All
3. POLARIZER MODE: TRANSMISSIVE/NORMALLY BLACK
4. DRIVER IC:ST7735S
5. OPERATING TEMP.: -20°C~70°C
6. STORAGE TEMP.: -30°C~80°C
7. BACK LIGHT: 1 CHIP-WHITE LED; 20MA,3.0V
8. LCM Luminance: 400 CD/M2(TYP)
9. UNMARKED TOLERANCE:±0.2
10. 建议机壳开窗可视区比 LCD A.A区单边大 0.3mm
11. 产品符合ROHS标准

Customer Approval Signature	GoldenMorning Electronics Co., Ltd.						Drawing Number	Rev.
							0.96" TFT	V1
	Unless Otherwise Specified		Title	GMT169-04-WG			Material	
	Unit	mm					Soda Lime / Polyimide	
	General Roughness							
	Tolerance			Drawn	E.E.	Panel / E.	P.M.	
	Dimension	±0.3	By				Scale	Sheet	Size
	Angle	±1	Date				1:1	1 of 1	A3

3. Maximum Ratings（极限参数）

Parameter 参数	Symbol 符号	Min 最小	Max 最大	Unit 单位
Logic Supply Voltage 逻辑电源电压	IOVCC	-0.3	4.6	**V**
Analog Supply Voltage 模拟电源电压	VCC	-0.3	4.6	**V**
Operating temperature 工作温度	Top	-20	70	℃
Storage temperature 储存温度	Tst	-30	80	℃
Humidity 湿度	RH	--	90%(Max60C)	RH

4. ELECTRICAL CHARACTERISTICS（电气特性）

Parameter 参数	Symbol 符号	Min 最小	Typ 典型	Max 最大	Unit 单位
Logic Supply Voltage 逻辑电源电压	IOVCC	1.65	1.8/2.8	3.3	V
Analog Supply Voltage 模拟电源电压	VCC	2.6	2.8	3.3	--
Input Current 输入电流	Idd	--	20	--	mA

5. BACKLIGHT CHARACTERISTICS（背光特性）

Item 项目	Symbol 符号	Min 最小	Typ 典型	Max 最大	Unit 单位	Condition 条件
Forward Voltage 正向电压	Vf	3.0	3.2	3.4	V	--
Forward Current 正向电流	If	--	20	--	mA	--
Operating Life Time 使用寿命	--	--	10000	--	Hrs	

Note 1: The LED Supply Voltage is defined by the number of LED at Ta=25℃
(LED 电源电压由 Ta=25℃时的 LED 数量定义)

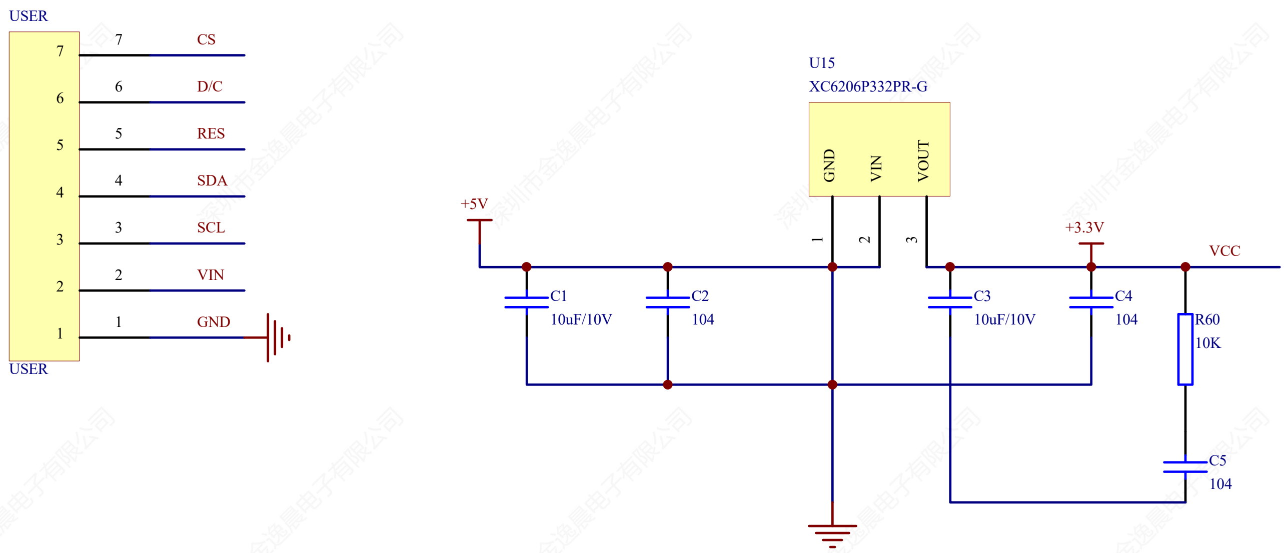
Note 2: Operating life means brightness goes down to 50% initial brightness. Typical operating life time is estimated data.
(使用寿命意味着亮度降低到初始亮度的 50%。典型的使用寿命是估算数据。)

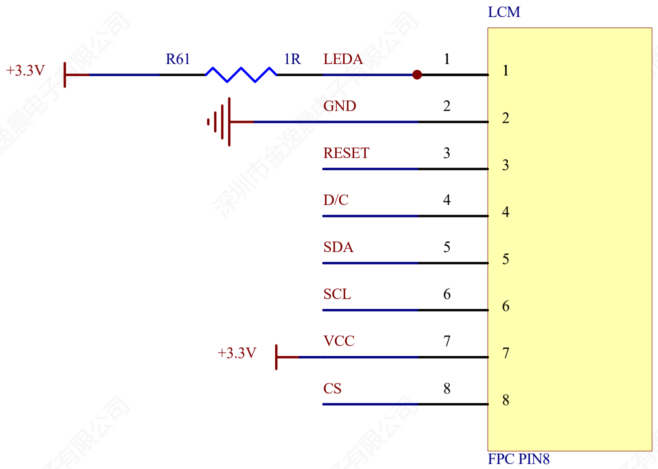
6. PIN DESCRIPTIONS（PIN 定义）

Pin.No 编号	Symbol 符号	Description 说明
1	LEDA	Anode of Backlight （背光正极供电脚）
2	GND	Ground （接地脚）
3	RESET	LCM Reset pin （屏复位脚）
4	D/C	Register select pin（指令/数据寄存器选择脚） D/C='1': Display data.（D/C='1':选择数据寄存器） D/C='0': Command data.（D/C='0':选择指令寄存器）
5	SDA	Serial data input (串口数据输入脚)
6	SCL	Serial clock pin. (串口时钟线)
7	VDD	Power supply for Analog（2.8V-3.3V） （系统电压）
8	CS	Chip select pin("Low" enable) （屏驱动芯片片选脚，低电平有效）

NOTE: The backlight LED can be powered separately or share a set of voltage supply with the VCC.
背光 LED 可以单独供电，也可以和 VCC 共用一组电压供电）

7. Schematic Diagram（原理图）





8. OPTICAL CHARACTERISTICS（光学特性）

Item	Symbol	Measuring Conditions		Min.	Typ.	Max.	Unit	Remark
Response Time	Tr+Tf	$\theta = 0^{\circ}$ $\phi = 0^{\circ}$	25 °C	-	30	40	ms	
Viewing Angle	θ	$\phi = 0^{\circ}$	25 °C	-	80	-	Deg	
		$\phi = 180^{\circ}$	25 °C	-	80	-		
	θ	$\phi = 90^{\circ}$	25 °C	-	80	-		
		$\phi = 270^{\circ}$	25 °C	-	80	-		
Contrast Ratio	CR	-	25 °C		800	-	-	
Color of CIE Coordinate	White	X	25 °C	0.304	0.306	0.308	-	-
		Y	25 °C	0.325	0.327	0.329		
	Red	X	25 °C	0.608	0.610	0.612		
		Y	25 °C	0.331	0.333	0.335		
	Green	X	25 °C	0.279	0.281	0.283		
		Y	25 °C	0.531	0.533	0.535		
	Blue	X	25 °C	0.144	0.146	0.148		
		Y	25 °C	0.136	0.138	0.140		
Transmittance (with polarizer)					5.09		%	

Note1:Definition of Response Time.(white-black)

The response time is defined as the time interval between the 10% and 90% amplitudes

9 Reliability（可靠性）

9.1 MTBF(平均故障间隔时间)

The LCD module shall be designed to meet a minimum MTBF value of 50000 hours with normal.
LCD 模块的设计应满足正常情况下 50000 小时的最小 MTBF 值。

9.2 Test Condition(测试条件)

No	ITEM 测试项目	CONDITION 条件	CRITERION 标准
1	High Temperature Non-Operating Test 高温非操作试验	80℃*240Hrs	.No Defect Of Operational Function In Room Temperature Are Allowable (室温下不允许有操作功能缺陷) .IDD of LCM in Pre-and Post-Test Should Follow Specification (LCM 在测试前和测试后的 IDD 应遵循规范)
2	Low Temperature Non-Operating Test 低温非操作试验	-30℃*240Hrs	
3	High Temperature/Humidity Non Operating Test 高温/高湿非操作试验	60℃*90%RH*240Hrs	
4	High Temperature Operating Test 高温运行试验	70℃*240Hrs	
5	Low Temperature Operating Test 低温运行试验	-20℃*240Hrs	
6	Thermal Shock Test 热冲击试验	-20 ℃(30Min)<>70 ℃(30Min) *10CYCLES	

Notes:

1. Judgments should be made after exposure in room temperature for two hours.
   应在室温下暴露两小时后做出判断。
2. The distill water is used for the high temperature/humidity test.
   蒸馏水用于高温/湿度试验。
3. The sample above is individually for every reliability tests condition.
   上面的样本是针对每种可靠性测试条件单独提供的。

10. Precautions（注意事项）

10.1 Storage Conditions（储存条件）

(1) Store the panel or module in a dark place where the temperature is 23±5℃and the humidity is below 45±20%RH.(将面板或模块存放在温度为 23±5℃、湿度低于 45±20%RH 的黑暗处)

(2) Store in anti-static electricity container.(储存在防静电容器中)

(3) Store in clean environment, free from dust, active gas, and solvent.(储存在清洁的环境中，没有灰尘、活性气体和溶剂)

(4) Do not place the module near organics solvents or corrosive gases.(请勿将模块放置在有机溶剂或腐蚀性气体附近)

(5) Do not crush, shake, or jolt the module.(请勿挤压、摇晃或震动模块)

10.2 Handling Precautions(处理注意事项)

(1) Avoid static electricity, which can damage the CMOS LSI.（避免静电，因为静电会损坏 CMOS LSI）

(2) The polarizing plate of the display is very fragile, please handle if very carefully. (LCM 上的偏振片非常脆弱，请小心处理)

(3)Do not give external shock.(不要进行外部电击)

(4) Do not apply excessive force on the surface.(不要在表面上施加过大的力)

(5) Do not wipe the polarizing plate with a dry cloth, as it may easily scratch the surface of plate.(不要用干布擦拭偏振片，这样很容易划伤偏振片表面)

(6) Do not operate it above the absolute maximum rating. （请勿在极限参数以上操作）

(8)Do not remove the panel or frame from the module.（请勿从模块上拆下面板或框架）